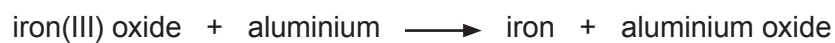


4. (a) When a mixture of iron(III) oxide and aluminium powder is heated, the following reaction takes place.



The reaction is commonly called the thermite reaction. The photographs show the reaction taking place and how it is used in the repair of railway lines.



- (i) Explain why iron is formed during the reaction.

[2]

.....

.....

.....

- (ii) Complete and balance the equation for the reaction.

[2]



- (iii) Calculate the percentage by mass of iron in iron(III) oxide, Fe_2O_3 .

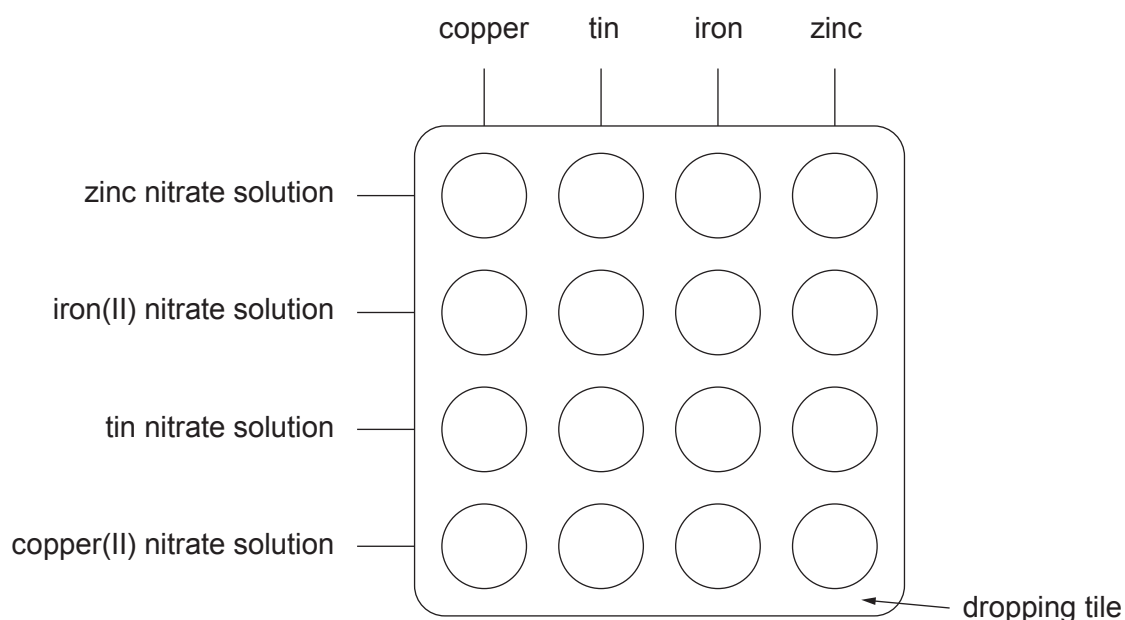
[2]

$$A_r(\text{Fe}) = 56 \qquad A_r(\text{O}) = 16$$

Percentage = %



- (b) Clare and Frankie were investigating the reactivity of metals. They carried out a series of displacement reactions in a dropping tile. In each test they placed a small piece of metal into a solution of the nitrate of a different metal as shown.



- (i) It was not necessary to carry out all of the tests. Place crosses (×) on the diagram to show which tests did not need to be carried out.

Explain your choice.

[2]

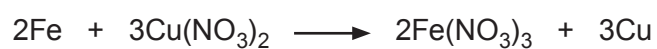
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- (ii) The equation shows the reaction between iron and copper(II) nitrate solution.



Use the equation to calculate the maximum mass of copper that you would expect to be formed when 0.224 g of iron is added to excess copper(II) nitrate solution.

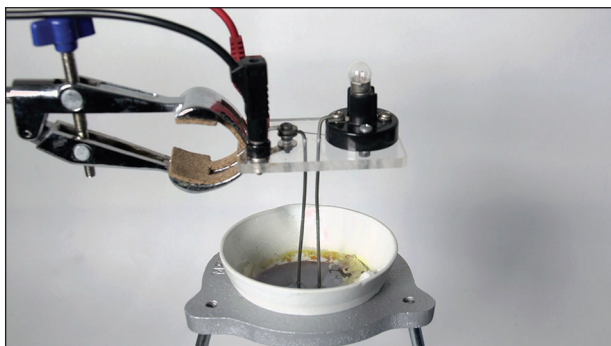
[3]

$$A_r(\text{Fe}) = 56 \quad A_r(\text{Cu}) = 63.5$$

Maximum mass of copper = g



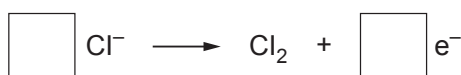
- (c) The photograph shows how the electrolysis of zinc chloride can be carried out in the laboratory.



- (i) Balance the equation that represents the reaction taking place at the anode.

Use the equation to explain the meaning of the term *oxidation*.

[2]



- (ii) The following method can be used to calculate the mass of zinc produced during the process.

- Record the mass of the cathode before placing it into the electrolyte
- Allow the process to run to completion
- Remove the cathode from the electrolyte and record its new mass
- Calculate the increase in mass

- I. It is often found that the increase in mass measured using this method is greater than expected. Suggest a reason for this.

[1]

- II. Suggest why this method cannot be used to measure the mass of chlorine produced during the process.

[1]

